

The Schur- c_0 problem for $X_{\mathcal{F}}$ is answered by the rapid Farah family

Literature-status note

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Status. `literature_already_answered`.

Source problem. Borodulin-Nadzieja and Farkas, *Analytic P -ideals and Banach spaces*, arXiv:1905.13484, ask in the section “Schur Property”, Problem labeled `schur-c0`:

Is there a family $\mathcal{F} \subseteq [\omega]^{<\omega}$ such that $X_{\mathcal{F}}$ does not have the Schur property

and $X_{\mathcal{F}}$ does not contain a copy of c_0 ?

Supporting answer. Borodulin-Nadzieja, Farkas, Jachimek, and Pelczar-Barwacz, *The Zoo of combinatorial Banach spaces*, arXiv:2404.01733, explicitly studies the same implication chain for combinatorial Banach spaces:

Schur property $\Rightarrow \ell_1$ -saturation $\Rightarrow$ no copies of c_0 .

In the introduction they state that none of these implications can be reversed. In Subsection “The rapid Farah family” they define a family `rFh`, note that the block intervals (I_n) witness that X_{rFh} does not have the Schur property, and prove in Theorem `e1e1` that X_{rFh} is ℓ_1 -saturated.

Why this answers the source problem. Take $\mathcal{F} = \text{rFh}$. The supporting paper gives directly that $X_{\mathcal{F}}$ is not Schur. It also proves $X_{\mathcal{F}}$ is ℓ_1 -saturated. Since a copy of c_0 has subspaces containing no copy of ℓ_1 (indeed c_0 itself contains no ℓ_1), an ℓ_1 -saturated space cannot contain c_0 . Therefore X_{rFh} is exactly the family requested in the 2019 problem.

Provenance and scope. This is a later-paper answer, not a new result from this run. The 2024 paper cites the 2019 paper as the source of the $X_{\mathcal{F}}$ framework and explicitly frames the rapid Farah construction as a non-reversal of the Schur-to- ℓ_1 -saturation implication. The packet answers the existence question only; broader classification or characterization questions for arbitrary $\mathcal{F}$ remain outside this status note.

References.

- P. Borodulin-Nadzieja and B. Farkas, *Analytic P -ideals and Banach spaces*, arXiv:1905.13484.
- P. Borodulin-Nadzieja, B. Farkas, S. Jachimek, and A. Pelczar-Barwacz, *The Zoo of combinatorial Banach spaces*, arXiv:2404.01733.